Week1 Report - Yun Chen

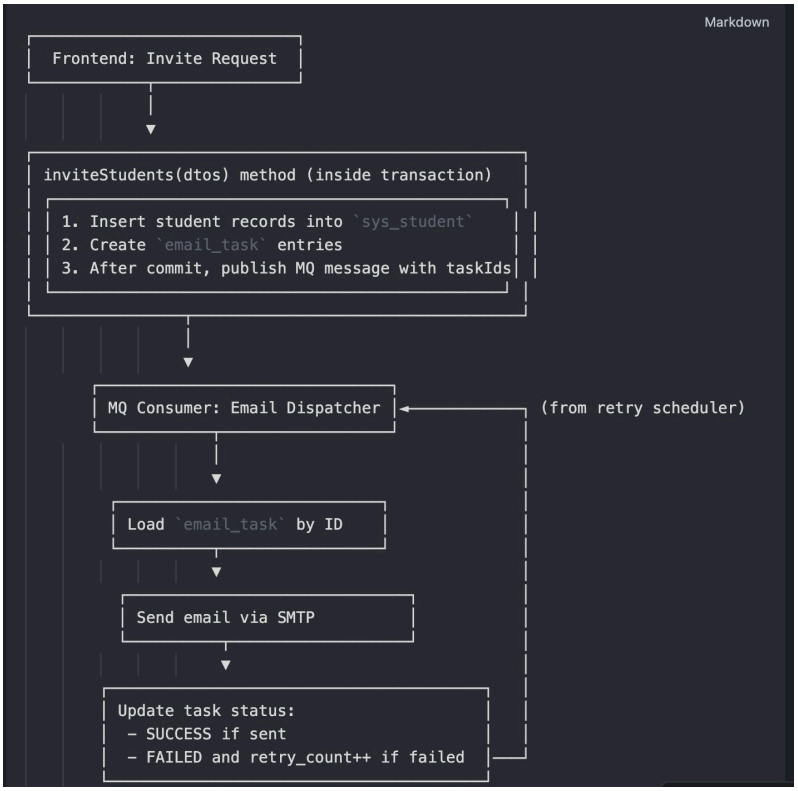
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Date: 1/8/2025

1. Work Completed

Since I had multiple interviews this week, I didn't do much coding. Instead, I focused on designing a solution for a legacy issue from last semester.

In the registration process, to prevent a situation where student information is saved to the database but email sending fails, I initially placed both operations within the same Spring transaction. However, this led to a new problem: since email sending is an external service, putting both actions in a single transaction could result in a rollback if email sending fails—even after the database insert has succeeded—while previously sent emails cannot be undone. This causes inconsistencies between the database and the external state. To solve this issue, I decided to adopt a "**message queue + email task table**" approach: email tasks are inserted into the database along with student data within the same transaction, and emails are sent asynchronously after the transaction is committed, ensuring eventual consistency and supporting failure retries and compensation.

New Workflow



New table structure

```sql
CREATE TABLE email_task (
    id BIGINT PRIMARY KEY AUTO_INCREMENT,
    student_id BIGINT NOT NULL,                        -- Foreign key to student
    to_address VARCHAR(255) NOT NULL,                  -- Target email address
    subject VARCHAR(255) NOT NULL,                     -- Email subject
    content TEXT NOT NULL,                             -- Email body
    status ENUM('PENDING', 'SUCCESS', 'FAILED') DEFAULT 'PENDING', -- Send status
    retry_count INT DEFAULT 0,                         -- Retry attempts
    max_retry INT DEFAULT 3,                           -- Max retry limit
    last_attempt_time TIMESTAMP NULL,                  -- Last send attempt
    create_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP  -- Task creation time
);
```

total 1.5h

2. Next Week Plan

- **start coding**